EarthPortal

From the Metadata Guidelines to curation actions

Auditing the semantic artefacts of EarthPortal with your guidelines and O'FAIRe

Emmanuel Nyandu Kagarabi

Internship at CNRS / Data Terra, 1 September to 31 October 2026 Supervisor: Christelle Pierkot

OntoPortal Federation Meeting, 13 November 2026

CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE

DATA TERRA

EarthPortal

SISSA

Today: 35 minutes of talk, then discussion

1. The subject and our goal
2. Your guidelines, machine-readable
3. O'FAIRe in five minutes
4. The pipeline: from the API to one audit per artefact
5. What we got on 10 November 2026
6. Reproducibility, and reuse on other portals
7. **Discussion**

The message of this talk

Your Metadata Guidelines became a **table that a program can read**. With it, and with O'FAIRe as a measuring instrument, every artefact of EarthPortal now has an **audit**: what is missing, what it costs, and who can fix it.

Everything shown today is open and citable: SA_Management v1.0.0, doi.org/10.5281/zenodo.23417011.

1

The subject and our goal

Where we started

92

entries in EarthPortal

68 artefacts, 19 views,
5 records without content

99

properties in your guidelines

20 Mandatory, 58 Recommended,
21 Optional

61

questions in O'FAIRe

478 points in total,
386 reachable on EarthPortal

Two references, one catalogue

The guidelines say **what a good description contains**; O'FAIRe **scores** how FAIR an artefact is from its description. EarthPortal stores the descriptions in **130 fields** (110 per submission, 20 per ontology record). Our work connects the three.

Snapshot of 10 November 2026. Guidelines: review version of September 2026, the base of our cheatsheet (Report 1).

Our goal

For each artefact of EarthPortal, answer three questions:

1. **What is missing** in its description, according to the guidelines?
2. **What does it cost** in FAIRness, according to O'FAIRe?
3. **Who can fix it**: the author, the curator, or the portal?

And do it **again and again**: each run is a dated snapshot, so we can follow the catalogue over time.

What we do not do

We do not change O'FAIRe. We use it **as an instrument**. But to read its scores correctly, we read its Java code (`agroportal/fairness`, commit `24def6b`).

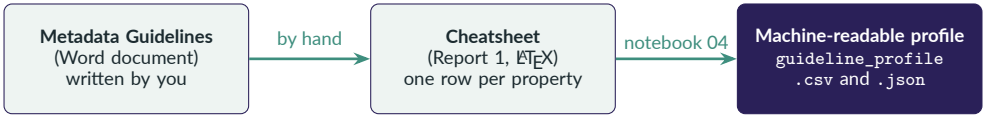
Who it is for

Authors (their audit), curators (the actions), and you: what the catalogue says about the guidelines.

2

Your guidelines, machine-readable

From a document to a machine-readable table



One row of the profile (property 1.9):

id, section	1.9, General
entry, usage	Version IRI, Recommended
property (guidelines)	`omv:versionIRI`
stored by EarthPortal as	`owl:versionIRI` (submission)
link rule	manual
who fixes it	author

Check out `guideline_profile.csv`.

What a program now knows, for all 99 properties:

- its usage and FAIR principle
- where the portal stores it (record or submission, API key)
- its allowed values, when the portal has a list
- who can fill it: author 86, curator 8, portal 5

The split author / curator / portal is our proposal, to be validated.

Guidelines and portal schema: 99 / 99 linked

Rule	Meaning	Properties
same	same name	72
mapping	a declared synonym	21
manual	linked by hand	6

A synonym is declared in the portal schema (`metadataMappings`): for example `dcat:keyword` is stored as `omv:keywords`. The 21 mapped properties need nothing.

Five names of the guidelines that the portal does not recognise

Id	Guidelines	EarthPortal field	Note
1.1	`mod:acronym`	`omv:acronym` (ontology record)	no synonym declared
1.9	`omv:versionIRI`	`owl:versionIRI`	MOD 3.2 uses `owl:versionIRI`: the guidelines may be the one to correct
10.1	`dct:isPartOf`	`viewOf` (ontology record)	the portal only has the inverse, `dct:hasPart`
10.5	`mod:semanticArtefactRelation`	`door:ontologyRelatedTo`	equivalent in MOD 3.2, but not listed by the portal
10.10	`mod:hasDisparateModelling`	`door:hasDisparateModelling`	the portal lists `mod:hasDisparateModellingWith`

If an author writes these names in the file, we expect the portal not to read them.

Report 5 (October 2026). Portal schema: `/submission_metadata`, `/ontology_metadata`; MOD:FAIR-IMPACT/MOD/`mod.ttl`.

Your question of 13 October

EarthPortal field (label)	Read by O'FAIRe
`omv:version` (version information)	R1.2Q4
`owl:deprecated` (deprecated)	A2Q4
`prefLabelProperty` (preferred label)	I3Q3, R1Q1
`synonymProperty` (synonym)	I3Q3, R1Q1
`definitionProperty` (definition)	I3Q3, R1Q1
`authorProperty` (author)	I3Q3, R1Q1
`obsoleteProperty` (obsolete)	I3Q3, R1Q1
`hierarchyProperty` (hierarchy)	I3Q3, R1Q2
`createdProperty` (creation date)	no
`modifiedProperty` (modification date)	no

The last eight are the "object description properties" of the portal.

Fields read by O'FAIRe, absent from the guidelines. 10 fields of the schema are filled from the file, but no property of the guidelines points to them; **8** of them bring O'FAIRe points.

Most values are defaults

In about 55 artefacts they hold the same values. The OntoPortal code gives defaults: `skos:prefLabel` (even for OWL), `owl:subClassOf` for OWL hierarchies, `dc:created`. As far as we know, the last two do not exist in their vocabularies.

Report 5 (October 2026)

3

O’FAIRe in five minutes

What is O'FAIRe?

O'FAIRe (*Ontology FAIRness Evaluator*) scores how FAIR a semantic artefact is, using **its metadata** on an OntoPortal instance.

- 4 principles → 15 sub-principles → **61 questions**
- each question pays credits, from 0 to its maximum m_q
- no human judgement: the same description always gets the same score

Principle	Questions	Maximum
Findable	13	113
Accessible	13	113
Interoperable	15	109
Reusable	20	143
Total	**61**	**478**

The score of an artefact o

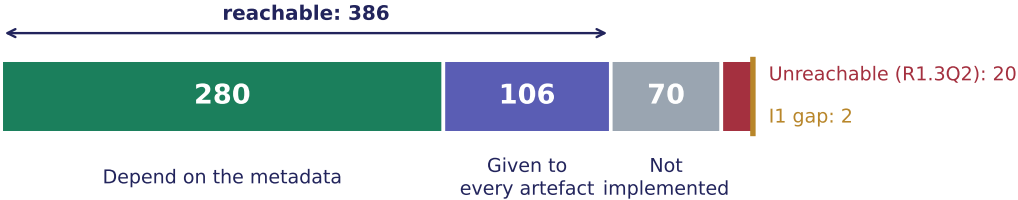
$$S(o) = \sum_{q=1}^{61} s_q(o), \qquad 0 \le s_q(o) \le m_q$$

$$N(o) = \left\lfloor 100 \cdot \frac{S(o)}{478} \right\rfloor$$

Example, OSO: $S = 364.36$, so $N = \lfloor 76.2 \rfloor = 76$.

Amdouni, Bouazzouni & Jonquet (2022), *O'FAIRe makes you an offer*, IJMSO 16(1).

What the 478 points measure on EarthPortal



No EarthPortal artefact can earn 92 of the 478 points: 10 questions are not implemented (70), R1.3Q2 rewards AgroPortal and OBO groups (20), I1 counts 44 for 42 (2). So, beside O'FAIRe's score, we report a **reachable score**:

$$R(o) = 100 \cdot \frac{S(o)}{386}$$

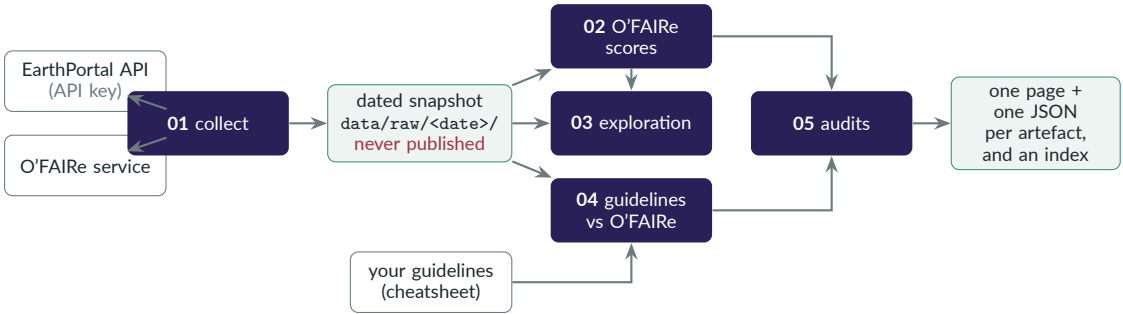
OSO	O'FAIRe *N*	Reachable *R*
	76 / 100	**94.4 %**

Read from the Java code (commit `24def6b`). The ranking of the artefacts is the same with both scores; only the scale changes.

4

The pipeline

Five notebooks: collect once, then compute



01	downloads the catalogue, the portal schema, the latest submission and the O'FAIRe result of every entry
02–03	turn O'FAIRe's answers into one table, and explore it
04	reads your guidelines, finds where the portal stores each property, and links each question to the properties it reads
05	writes the audits

How a lost point becomes an action



Every **empty** property of the guidelines is an action. When a question loses L_q points and reads n_q empty properties, each of them gets a share:

$$\text{share}(a) = \sum_{q \text{ reads } a} \frac{L_q}{n_q}$$

A way to rank actions, not a promise of points.

Priorities: guidelines first, then points

P1	Mandatory
P2	Recommended, costs O'FAIRe points
P3	Optional, costs O'FAIRe points
P4	Recommended
P5	Optional

One audit per artefact: 1. FAIRness

SWEET: SWEET Ontology

Kind: **artefact** · Snapshot: 2026-11-10 · Portal page: https://earthportal.eu/ontologies/SWEET

This audit has two parts: the FAIRness of the ontology, as measured by O'FAIRe, and its compliance with the Metadata Guideline of the OntoPortal Federation. The suggested actions combine both.

1. FAIRness (O'FAIRe)

O'FAIRe score	Points	Reachable score	Rank
58 / 100	281.75 / 478	73 % of 386	11 of 68 artefacts

Where O'FAIRe points are lost

Question	Points	Lost	O'FAIRe says
F1Q2	0 / 11	11	External identifier is absent.
F1Q4	0 / 9	9	Version URI is absent.
F4Q2	4 / 10	6	The ontology is registered in 2 ontology repositories.
A1Q1	0 / 6	6	The ontology provides 0 resolvable identifier.
A1Q4	0 / 6	6	The ontology is not accessible through a SPARQL endpoint.
R1.2Q2	0 / 6	6	The ontology provides 0 property to inform on its general provenance.
R1.2Q3	0 / 6	6	The ontology provides 0 accrual property.
R1.2Q6	0 / 6	6	The ontology provides 0 property to inform about methodology and tools used.

earthportal_sa_audit/audits/SWEET.md, rendered by GitHub (snapshot of 10 November 2026).

Example: **SWEET**, the 11th artefact of the ranking.

Its scores, then every question where it loses points, largest loss first, with **O'FAIRe's own sentence**.

One audit per artefact: 2. Your guidelines

2. Compliance with the Metadata Guideline

Filled means that the portal has a value for the property; whether the value is correct is not checked here.

Usage	Filled	Empty	Not applicable	Total
Mandatory	16	4	0	20
Recommended	17	40	1	58
Optional	3	17	1	21

▼ The 99 properties of the guideline, one by one

Id	Property	Usage	Status
1.1	Acronym	Mandatory	filled
1.2	Name	Mandatory	filled
1.3	Representation language	Mandatory	filled
1.4	Categories	Mandatory	filled
1.5	Groups	Optional	filled

Not applicable: This ontology is a view of (only views) and *Was invalidated by* (only retired artefacts).

SWEET fills **16 of the 20** Mandatory properties, but only **17 of the 57** Recommended ones that apply.

The 99 properties are listed one by one in a folded block, each *filled*, *empty* or *not applicable*.

One audit per artefact: 3. Actions, and after

3. Suggested actions (SWEET, first rows)

3. Suggested actions

Each empty property of the guideline is an action. **P1**: Mandatory. **P2**: Recommended, and costing O'FAIRe points. **P3**: Optional, and costing O'FAIRe points. **P4**: Recommended. **P5**: Optional. *Points* is the share of the O'FAIRe points lost on the questions in brackets, divided among the empty properties these questions read.

By the author (in the file or the submission form)

Priority	Property	Guideline	O'FAIRe points	Stored as
P1	**Modification date** (3.2)	Mandatory	none	`omv:modificationDate`
P2	**Version IRI** (1.9)	Recommended	9 (F1Q4)	`owl:versionIRI`
P2	**SPARQL endpoint** (6.8)	Recommended	6 (A1Q4)	`sd:endpoint`
P2	**Has equivalences with** (10.7)	Recommended	5.1 (I2Q4, I3Q3)	`door:isAlignedTo`
P2	**Endorsed by** (5.7)	Recommended	5 (R1.3Q1)	`omv:endorsedBy`
P2	**Source** (6.9)	Recommended	3 (R1.2Q2)	`dct:source`
P2	**Is format of** (6.2)	Recommended	2.1 (I1Q5, I3Q3)	`dct:isFormatOf`
P2	**Has format** (6.3)	Recommended	2.1 (I1Q5, I3Q3)	`dct:hasFormat`
P2	**Funded or sponsored by** (5.6)	Recommended	2 (R1.2Q8)	`foaf:fundedBy`
P2	**Design for task** (9.2)	Recommended	2 (R1.2Q7)	`omv:designedForOntologyTask`
P2	**Created With** (12.3)	Recommended	2 (R1.2Q6)	`omv:usedOntologyEngineeringTool`
P2	**Accrual method** (12.4)	Recommended	2 (R1.2Q3)	`dct:accrualMethod`
P2	**Accrual periodicity** (12.5)	Recommended	2 (R1.2Q3)	`dct:accrualPeriodicity`
P2	**Accrual policy** (12.6)	Recommended	2 (R1.2Q3)	`dct:accrualPolicy`
P2	**Competency question** (12.7)	Recommended	2 (R1.2Q7)	`mod:competencyQuestion`
P2	**Contributor** (5.3)	Recommended	1.7 (R1.2Q1)	`omv:hasContributor`
P2	**Translator** (5.8)	Recommended	1.7 (R1.2Q1)	`schema:translator`
P2	**Similar to** (10.4)	Recommended	0.6 (I2Q6, I3Q3)	`door:similarTo`

Each row says **why the guidelines ask** for the property and **what it is worth** in O'FAIRe.

One JSON file per artefact

The same audit, as data: `audits/json/<ACRONYM>.json`. The page is written from it. It holds statuses, scores and actions, and no personal data.

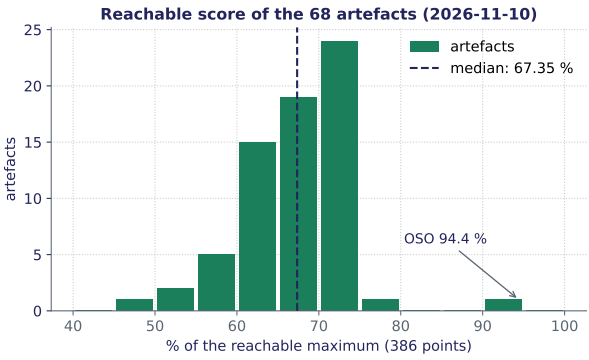
A curation interface

Amine Kasmi (EarthPortal) is building a curation interface on these JSON files: work in progress (Report 6).

5

What we got on 10 November 2026

How FAIR is the catalogue?



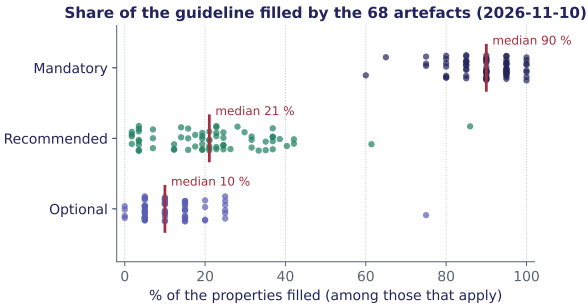
68 artefacts:

- reachable score: median **67.35 %**, from 49.5 % (STY) to 94.4 % (OSO)
- O'FAIRe's own score: median **54 / 100**
- one clear leader; most artefacts between 60 and 75 %

Between 16 October and 10 November

The status of the 99 properties is identical. Six scores moved only through O'FAIRe's live checks (for example SEWERNET -3.9, GEOLTSBOUND $+2.3$).

How much of your guidelines is filled?



Share of the properties filled, among those that apply, for each of the 68 artefacts:

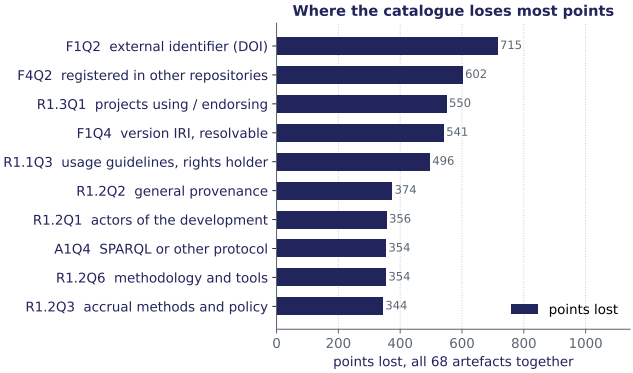
- **Mandatory**: median **90 %**
- **Recommended**: median **21 %**
- **Optional**: median **10 %**

Two stories

The median artefact reaches 67 % of what O'FAIRe can give, but fills only a fifth of what your guidelines recommend.

"Filled" means present; the value is not checked.

Where the catalogue loses most points



Points lost $= \sum_o (m_q - s_q(o))$ over the 68 artefacts.

- **F1Q2**: 65 artefacts have no external identifier; only a registered **DOI** gives the full 11 points
- **F4Q2**: EarthPortal is not in O'FAIRe's list of repositories
- **R1.3Q1**: 50 artefacts say nothing about projects using them or organisations endorsing them

Other identifier (a DOI) is the action with the largest share for 57 of the 68 artefacts.

Who can act? Mostly the authors

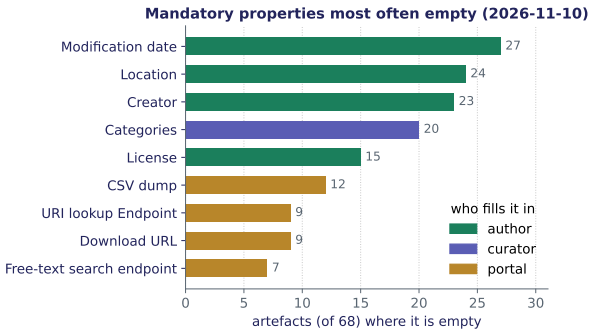
4 378 actions for the 68 artefacts:

Who	P1 to P3	All (P1 to P5)
author	2 626	4 163
curator	81	178
portal	37	37

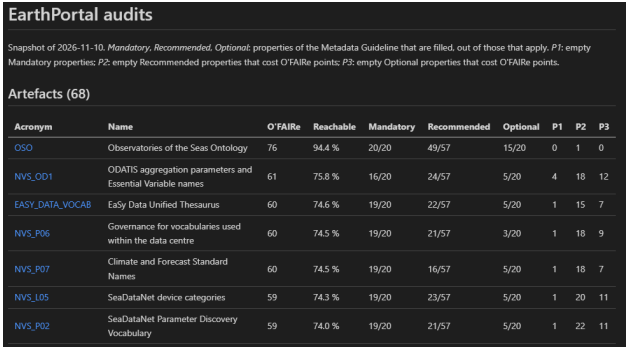
Preventing is better than curing

The submission form does not ask for the **licence**, the **creator** or the **modification date**, and **categories** are not required: four of the five Mandatory properties most often empty.

Mandatory properties most often empty



The whole catalogue on one page



EarthPortal audits

Snapshot of 2026-11-10. *Mandatory*. *Recommended*. *Optional*: properties of the Metadata Guideline that are filled, out of those that apply. *P1*: empty Mandatory properties; *P2*: empty Recommended properties that cost O'FAIRe points; *P3*: empty Optional properties that cost O'FAIRe points.

Artefacts (68)

Acronym	Name	O'FAIRe	Reachable	Mandatory	Recommended	Optional	P1	P2	P3
OSO	Observatories of the Seas Ontology	76	94.4 %	20/20	49/57	15/20	0	1	0
NVS_OD1	ODATIS aggregation parameters and Essential Variable names	61	75.8 %	16/20	24/57	5/20	4	18	12
EASY_DATA_VOCAB	EaSy Data Unified Thesaurus	60	74.6 %	19/20	22/57	5/20	1	15	7
NVS_P06	Governance for vocabularies used within the data centre	60	74.5 %	19/20	21/57	3/20	1	18	9
NVS_P07	Climate and Forecast Standard Names	60	74.5 %	19/20	16/57	5/20	1	18	7
NVS_L05	SeaDataNet device categories	59	74.3 %	19/20	23/57	5/20	1	20	11
NVS_P02	SeaDataNet Parameter Discovery Vocabulary	59	74.0 %	19/20	21/57	5/20	1	22	11

`audits/index.md`: every entry, best reachable score first, with its guideline properties filled and its P1 to P3 actions; each acronym opens its audit.

Seen in the data

Two groups of artefacts give exactly the same answers to every question: DIVCHRONOSTRAT, DIVGEOCHRONO, GEOLTSBOUND, GEOLUNITTYPE; and TERRA_VOCABS_ESS, TERRA_VOCABS_ESV. Probably the same description template: worth a curator's look.

6

Reproducibility and reuse

Open, reproducible and citable



github.com/EarthPortal/SA_Management
MIT licence

- **Dated snapshots.** Only notebook 01 uses the network; 02 to 05 always give the same results for the same date.
- **One setup.** Python 3.12+, one `requirements.txt`, one `.env` with your API key.
- **Explained for non-programmers.** Each notebook block comes with a plain-text explanation.
- **Citable.** Each GitHub release is archived on Zenodo with a DOI:
  software 10.5281/zenodo.23417011 (v1.0.0), slides 10.5281/zenodo.23391527.
- **Documented.** `ARCHITECTURE.md` (structure), six short reports in `docs/`.

Running the pipeline on another portal

What stays the same

- your guidelines (the cheatsheet) and the machine-readable profile
- the five notebooks and the audit format
- the reading of O'FAIRe

What to change or check

1. the portal's API, web address and O'FAIRe service in `.env`
2. an API key of that portal
3. the 6 manual links, if its schema differs
4. the reachable maximum (386), which depends on the portal

Step by step in `ADAPTING.md`, at the root of the repository.

Limits, said plainly

Presence, not correctness

A property counts as filled when it is not empty. Default values and wrong IRIs pass, in O'FAIRe and in our audits.

O'FAIRe on the server

We read commit `24def6b`; we do not know which version runs on EarthPortal, nor how often its results are refreshed.

Who fixes it

The split author / curator / portal is a proposal. No audit has been sent to an author yet.

The guidelines move

Our cheatsheet follows the review version of September 2026. A new version of the guidelines means a new cheatsheet, then a new run.

Acknowledgements

Christelle Pierkot

My supervisor, whose guidance and support made this pipeline, and its latest results, possible.

Clément Jonquet

For O'FAIRe, and for spending a day with us at Data Terra to discuss it and the way we use it in this pipeline.

The Editorial Board

For welcoming me into your team, and for the Metadata Guidelines: the cornerstone (*la pierre angulaire*) of this pipeline.

The Data Terra team

Christelle, Nicolas, Hakim, Imene, Amirat, and Amine: a welcoming, supportive and dynamic team. It was a great honour and pleasure to work with you.

EarthPortal

Thank you

Merci beaucoup · Vielen Dank
Grazie mille

"All models are wrong, but some are useful."
George E. P. Box

"Big goals, small steps."

Software
10.5281/zenodo.23417011

These slides
10.5281/zenodo.24145898

CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE

DATA TERRA

EarthPortal

SISSA

Let’s discuss

About the guidelines

1. The **five names** the portal does not recognise: change the guidelines, or declare new mappings in the portal?
2. Add the **eight fields** O’FAIRe reads (version information, deprecated, object description properties), with their allowed values?
3. Could the **machine-readable profile** be published and versioned with the guidelines?

About the portals and O’FAIRe

4. Should the submission form ask for the **licence**, the **creator**, the **modification date**, and require **categories**?
5. Should **Location** stay Mandatory when a file is uploaded?
6. Could EarthPortal, and the other portals of the Federation, be added to O’FAIRe’s list of **repositories** (F4)?

Backup slides

The ten questions O'FAIRe does not implement

Question	Max	Text
F4Q3	8	Are the ontology libraries or repositories properly indexed by Web search engines?
I2Q3	3	If yes, does it include the minimum information for those terms?
I2Q5	7	If yes, are those alignments well represented and to unambiguous entities? …
I3Q1	20	Does an ontology provide qualified cross-references to external resources/databases?
I3Q2	6	If yes, are those cross-references well represented and to unambiguous entities?
R1Q3	7	How much of the ontology objects are described with labels?
R1Q4	6	How much of the ontology objects are defined using a text description?
R1Q5	6	How much ontology objects are defined using a property restriction or an equivalent class?
R1Q6	5	How much of the ontology objects provide provenance information with annotation properties?
R1.2Q5	2	Are the ontology latest changes documented?
Total	**70**	Every answer: "This question cannot be evaluated in the repository at this time."

F4: the repositories O'FAIRe knows

F4Q1 and F4Q2 count the catalogues of `includedInDataCatalog` that appear in O'FAIRe's list (`catalogs.config.json`, commit `24def6b`): **23 repositories** and **13 libraries**.

Repositories	NCBO BioPortal, AgroPortal, SIFR BioPortal, EBI OLS, Ontobee, OntoHub, AberOWL, LOV, ONKI, MMI ORR, ESIP COR, LifeWatch EcoPortal, MatPortal, MedPortal, CISMeF HeTOP, FINTO, ANDC RVA, NVS, GFBIO TS, Loterre, Caliper, Crop Ontology Curation Tool, Planteome
Libraries	OBO Foundry, Agrisemantics, FAIRsharing, ROMULUS, DAML Ontology Library, Colore, BARTOC, TaxoBank, LusTRE, LOV4IoT, VOCAB OEG, LiveSchema, Protégé Ontology Library

EarthPortal is not in the list, so an artefact hosted on EarthPortal does not get the point of "its own" portal (F4Q1), and the catalogue loses 602 points on F4Q2.

Who fixes what? The split author / curator / portal

III. The machine-readable guideline

We add two pieces of information that a program needs: what kind of value the portal expects (an IRI, a list, a date...) with its allowed values, and **who can fill the field in**. The last one is a proposal, to be validated with the team:

- **portal**: the portal generates the value itself (download links, search endpoints, submission date);
- **curator**: the EarthPortal team sets it (categories, groups, visibility, administrators, curation);
- **author**: everything else, filled in the file or in the submission form by the producer.

```python
PORTAL = {"bpm:csvDump", "void:dataDump", "void:uriLookupEndpoint", "void:openSearchDescription", "omv:creationDate"}
CURATOR = {"bpm:administeredBy", "bpm:group", "bpm:viewingRestriction", "bpm:viewOf",
           "schema:includedInDataCatalog", "pav:curatedBy", "pav:curatedOn"}

def who_fixes(row):
    if row.portal_attribute in PORTAL:
        return "portal"
    if row.portal_attribute in CURATOR or (row.entry == "Categories"):
        return "curator"
    return "author"
```

Notebook: `04_guideline_vs_ofaire.ipynb`.

The snapshots so far

Snapshot	Entries	Artefacts	Views	No content	Median reachable	OSO
28 Sept 2026	91	70	16	5	about 67 %	93.9 %
8 Oct 2026	92	68	19	5	67.85 %	94.4 %
16 Oct 2026	92	68	19	5	67.85 %	94.4 %
10 Nov 2026	92	68	19	5	67.35 %	94.4 %

- early October: `OSO-DCAT`, `OSO-VOID` and `OSO-SHACL` became views of OSO
- 10 November: **6 206 actions** for the 92 entries (P1 224, P2 2 812, P3 999, P4 1 517, P5 654)

`earthportal_sa_audit/data/ofaire_scores_<date>.csv`: one file per snapshot.